

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
A-LEVEL · FURTHER MATHEMATICS

MATHEMATICS 9187/2

PAPER 2 November 2023

3 hours

PAPER 2 — STRUCTURED QUESTIONS — CALCULATOR ALLOWED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2023 9187/2.

Additional materials: A calculator may be used unless a question forbids it.

INSTRUCTIONS: Answer all questions. Full marks are not given for answers only — show method.

1. A particle is projected at 20 m/s at 30° to the horizontal. Take $g = 10 \text{ m/s}^2$. Find time of flight on level ground and range. **[10]**
2. A 4 kg mass on a smooth plane inclined at 30° . Find acceleration down the plane ($g=10$). **[10]**
3. A uniform beam 4 m, mass 6 kg, supported at both ends. A 9 kg mass is 1 m from A. Find reactions ($g=10$). **[10]**
4. $X \sim B(5, 1/3)$. Find $P(X=2)$. **[8]**
5. $X \sim N(50, 16)$. Find $P(X > 54)$ in terms of Φ , using $Z=(X-\mu)/\sigma$. **[8]**
6. Calls arrive at 3 per hour. Using Poisson, write $P(\text{exactly 2 in an hour})$. **[8]**
7. Further Statistics: Hypothesis test outline. Give definitions and one calculation. **[8]**
8. Further Statistics: Correlation interpretation. Give definitions and one calculation. **[8]**
9. Further Statistics: Discrete random variable $E(X)$. Give definitions and one calculation. **[8]**
10. Further Statistics: Continuous pdf $f(x)=k$ on $[0,2]$. Give definitions and one calculation. **[8]**

11. Further Statistics: Sampling / CLT one-liner. Give definitions and one calculation. **[8]**

12. Further Statistics: Chi-square idea. Give definitions and one calculation. **[8]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).